A The Survey as Administered

The survey was a web application in three parts: onboarding, five stages and a completion step. The study information said the survey would take about 90 minutes and asked contributors not to use AI-generated text, because the study sought their own words. In Stages 1 to 4, contributors typed questions to the building into a chat box, which saved each question and gave no answer. Stage 5 had no chat.

Onboarding. A single screen held three steps. The first gave the study title (*A Survey-Based Study to Develop a Corpus of Natural Language Queries for Smart Building Interaction*), the ethics reference, a contact address and the information sheet to view or download. The second was a consent form of nine statements: that the contributor had read the information sheet; that taking part was voluntary and they could withdraw at any time with their data deleted; who could see the data and how it was protected; that anonymised quotations might be published; and that they agreed to take part. Contributors ticked every statement and typed their name, which produced a signed, time-stamped consent record. The third step opened only after consent and asked for a username (no password) and one or more of fourteen role profiles: Facility Managers / Building Maintenance Teams; Building Owners / Property Managers; Occupants / Tenants / Employees; Health and Safety Officers; Sustainability and Energy Management Teams; Compliance and Regulatory Bodies; Insurance Companies; Architects / Building Designers; Students / Researchers / Academics; IT / Data Scientists; Ontology Experts; Vendors / Service Providers; Real Estate Developers; Guests / Visitors. Contributors could leave and return later under the same username.

Personas. We count each contributor under the first profile they chose, shortened to Guests, Occupants, Students, Health & Safety Officers, Sustainability & Energy Teams, Building Owners, IT Professionals or Facility Managers. These eight are the only first choices: no contributor chose Compliance and Regulatory Bodies, Insurance Companies, Architects / Building Designers or Ontology Experts, and contributors chose Real Estate Developers and Vendors / Service Providers only after another profile.

Stages 1–4. Each stage set a target that the application encouraged but never required: clicking *Next* below the target showed the running count and an invitation to add more, and the contributor could continue regardless. The protocol asked for about 10 to 25 questions per stage. Most chat replies read "#N saved". At the tenth question, the chat said around 20 to 25 questions were wanted in the stage; at the twentieth, it suggested adding more.

Stage 1, *Open Minded.* Subtitle: "Ask anything — let your curiosity lead." Shown: a scenario only. Contributors were to imagine being inside a smart building (an office, university or public space) with a built-in AI that can answer anything about it, such as energy, temperature, air quality, occupancy and security, and to write what they would ask it, based on their own role and daily experience. No sensor list, examples or hints. Target: at least 15 questions.

Stage 2, *Sensing & Semantics.* Subtitle: "Discover what the building can sense." Shown: 43 sensing terms in eight groups. Environment: temperature, humidity, CO_2, VOC, PM2.5/PM10, ozone, NO_2. Occupancy: occupancy, motion, people count, door and window sensors. Lighting: lux, ambient light, glare index. Acoustics: sound level, noise events. Energy: kWh, kW, voltage, current, power factor, frequency. Utilities: gas, water, flow rate. HVAC: status, setpoint, supply and return air temperature, fan speed, chiller, boiler, pump. Other: air quality index, pressure, wind speed, vibration, equipment health, battery, smart plug, standby load. Clicking a term opened a plain-language description of what it measures, why it matters and how buildings use it, ending with a prompt to think of questions; contributors were told they need not open every term. Target: 15 terms explored and 20 questions.

Stage 3, *Scenario-Based.* Subtitle: "Explore real building spaces." Shown: eight spaces (open-plan office; meeting room; server room or data centre; lobby and reception; café and break area; rooftop and external plant; car park and EV charging; stairwells and corridors), each with a description, what is monitored there and three "picture yourself here" hints, for example "Picture yourself in a 2-hour meeting — how

would the air feel after the first hour?". In a three-dimensional view of Building A, the stage directed contributors to the fifth floor, where they could select a sensor and see a chart of its recorded readings. Contributors who had not opened the view saw a reminder but could still move on. Target: 5 spaces explored, 15 questions and the view opened at least once.

Stage 4, *Goal-Oriented*. Subtitle: "Sustainability & well-being goals." Shown: ten themes (energy efficiency, thermal comfort, lighting, indoor air quality, acoustic comfort, data analysis and insights, indoor transport, indoor pollution, water usage, land use and ecology), each with a description, three example ideas and keywords such as PMV/PPD, baseload, glare, VOCs and leak detection. Target: 4 themes explored and 10 questions or goals.

Stage 5. Rankings & Recommendations had no chat and three tasks, all required; the *Finish* button stayed locked until all three were complete. Of the 96 contributors, 80 ranked the topics, 69 ranked the example questions and 59 answered the open prompts (Appendix C).

Task 1, topics. Contributors ranked twenty topics by importance for future smart buildings, by clicking or dragging: Indoor Temperature Control; Air Quality & Ventilation; Lighting & Daylight; Noise & Acoustics; Energy Consumption; Security & Access Control; Occupancy & Space Utilisation; Fire Safety & Emergency; Water Management; Waste & Recycling; Renewable Energy & Solar; Green Spaces & Biodiversity; Health & Well-being; IoT Sensors & Data Analytics; Lifts, Stairs & Internal Transport; Parking & EV Charging; Building Automation & AI; User Apps & Digital Interaction; Building Maintenance & Faults; Carbon Footprint & Net Zero. The screen showed shorter names (for example *Temperature*).

Task 2, questions. For each of their ten highest-ranked topics, contributors ranked four example questions from most to least helpful. The instrument named and defined the four levels: *Basic*, a simple fact or current reading ("What is the current CO_2 level?"); *Intermediate*, comparisons across zones or spaces; *Advanced*, trends and patterns over time from sensor data; *Expert*, reports, visualisations, recommendations and alerts. For Indoor Temperature Control, the four questions were: "What is the current temperature reading in a specific room?" (Basic); "Compare temperature settings and readings across all zones in the building." (Intermediate); "Analyse temperature trends and daily patterns over the past month to detect anomalies." (Advanced); and "Generate a thermal comfort report with recommendations to improve temperature consistency across floors." (Expert).

Task 3, vision. Under the title *Your Vision for a Talking Building*, contributors imagined a ChatGPT-like building assistant that anyone could use, with or without technical knowledge, and answered five prompts in at least 20 words each: real scenarios and unmet needs (a time they wished they could ask a building something); interaction and response preferences (text, charts, alerts, dashboards or voice); trust, privacy and concerns (what would make them trust it, and what would stop them using it); success criteria and impact (what would make it useful for their role, and how to measure that); and current tools and pain points. An AI model proposed codes for the trust and success answers from a codebook, and the first author reviewed them one by one (§5.5); we report them in §6.

Completion. A confirmation prompt gave each contributor a unique reference number to submit on MTurk. The research team reviewed each contributor's answers to check they were genuine and not AI-generated. Contributors could rate the survey from one to five stars.

B Further Survey Analyses from the Automated Classification

Every result in this appendix that uses a question's domain, query type, intent, complexity, temporal scope or spatial scope describes the corpus as labelled by the keyword classifier (§3.5). Its agreement with a second automated classifier stayed below 0.70 on every dimension (Table 4). We report what answering the questions

Table 15. Leading labels on five of the six dimensions, from the automated classifier (agreement in Table 4), over all question submissions.

Dimension	Label	*n*	%
Domain	OTHER	1,467	20.5
	ENERGY	1,250	17.5
	SAFETY	763	10.7
	AIR_QUALITY	686	9.6
	THERMAL	594	8.3
Query type	STATUS	4,793	67.0
	CAPABILITY	1,704	23.8
	COMPARISON	175	2.5
	ANOMALY	160	2.2
Intent	INFORMATIONAL	6,558	91.7
	DIAGNOSTIC	280	3.9
	PRESCRIPTIVE	230	3.2
	PREDICTIVE	83	1.2
Complexity	LOOKUP	6,293	88.0
	AGGREGATION	610	8.5
	MULTI_STEP	248	3.5
Temporal scope	STATIC	6,429	89.9
	REALTIME	570	8.0
	PREDICTIVE	73	1.0

needs in §6, from the Part A codes, which an AI model proposed and the first author reviewed one by one (§5.5). The topic ranking and depth preferences (Tables 18 and 20) do not use the classifier: they are contributors' own rankings.

B.1 The classifier's labels

The classifier labels each submission on six dimensions. *Domain*: the twenty labels of Table 16. A question touching several domains receives the one its answer would mainly report on. `OTHER` marks any submission that matches no domain's keyword list or has fewer than 2 words or 3 characters. *Query type*: `STATUS`, `HISTORICAL`, `COMPARISON`, `ANOMALY`, `RECOMMENDATION`, `DIAGNOSTIC` or `CAPABILITY`. *Intent*: `INFORMATIONAL`, `DIAGNOSTIC`, `PRESCRIPTIVE` or `PREDICTIVE`. *Temporal scope*: `STATIC`, `REALTIME`, `HISTORICAL` or `PREDICTIVE`. *Spatial scope*: `POINT`, `ROOM`, `FLOOR`, `BUILDING`, `CAMPUS` or `UNSPECIFIED`. *Complexity*: `LOOKUP`, `AGGREGATION` or `MULTI_STEP`. The classifier partly derives complexity from query type: unless a complexity keyword matches first, it labels every `COMPARISON` and `ANOMALY` question `MULTI_STEP` and every `HISTORICAL` question `AGGREGATION`. When no keyword matches, it falls back to `STATIC`, `INFORMATIONAL` and `LOOKUP`.

B.2 The corpus by label

Table 15 gives the leading labels on five dimensions, and Table 16 every domain label. The twelve smallest domains together hold 1,483 questions (20.7%).

Table 16. All twenty domain labels, from the automated classifier (agreement in Table 4). OTHER marks a submission that matched no domain.

Domain	*n*	%	Domain	*n*	%
OTHER	1,467	20.51	WATER	213	2.98
ENERGY	1,250	17.48	WELLBEING	208	2.91
SAFETY	763	10.67	WAYFINDING	167	2.34
AIR_QUALITY	686	9.59	INFO_REQUEST	117	1.64
THERMAL	594	8.31	CONTROL	59	0.83
LIGHTING	334	4.67	TRANSPORT	56	0.78
SECURITY	312	4.36	PRIVACY	53	0.74
OCCUPANCY	262	3.66	ACCESSIBILITY	46	0.64
SUSTAINABILITY	258	3.61	WASTE	41	0.57
MAINTENANCE	224	3.13	WEATHER_OUTDOOR	41	0.57

Table 17. The four question-writing stages. A: number of questions, mean words per question, complexity labels and domain diversity (Shannon entropy of the stage's domain labels, natural logarithm). B: intent labels. C: share of the stage's questions in no domain, in energy and in air quality. Complexity, intent and domain are the automated classifier's labels (agreement in Table 4).

A. Questions

Stage	*n*	Mean words	Lookup %	Agg. %	Multi %	Diversity
S1 Open Minded	1,904	8.54	88.2	9.7	2.0	2.35
S2 Sensing & Semantics	2,003	10.03	85.8	9.5	4.6	2.31
S3 Scenario-Based	1,705	10.34	89.6	7.6	2.8	2.51
S4 Goal-Oriented	1,539	10.48	88.8	6.8	4.4	2.52

B. Intent labels (counts)					**C. Domain share (%)**	
Stage	**Diag.**	**Inform.**	**Pred.**	**Presc.**	**No domain**	**Energy / air**
S1	67	1,755	16	66	34.2	12.7 / 4.0
S2	115	1,836	21	31	13.7	23.3 / 17.7
S3	49	1,590	15	51	18.8	15.3 / 7.5
S4	49	1,377	31	82	14.3	18.3 / 8.4

B.3 The stages by label

Intent labels differ little across stages (Table 17; Cramér's $V = 0.060$): diagnostic questions were most frequent in Stage 2 and prescriptive ones in Stage 4. The χ^2 test behind V treats the submissions as independent, although each contributor wrote many, so we report its effect size without a p-value. Figure 17 gives the fifteen most distinctive terms of each stage, and Figure 18 the share of each stage's questions counted as novel and the complexity labels by stage (§3.4).

B.4 Domains by intent and by persona

Figure 19 shows the intent labels within each of the fourteen largest domains, and Figure 20 each persona's share of questions in the twelve largest. Informational questions dominate all fourteen domains shown. Of these, air quality has the largest diagnostic share (9.3% of its questions) and sustainability the largest prescriptive share

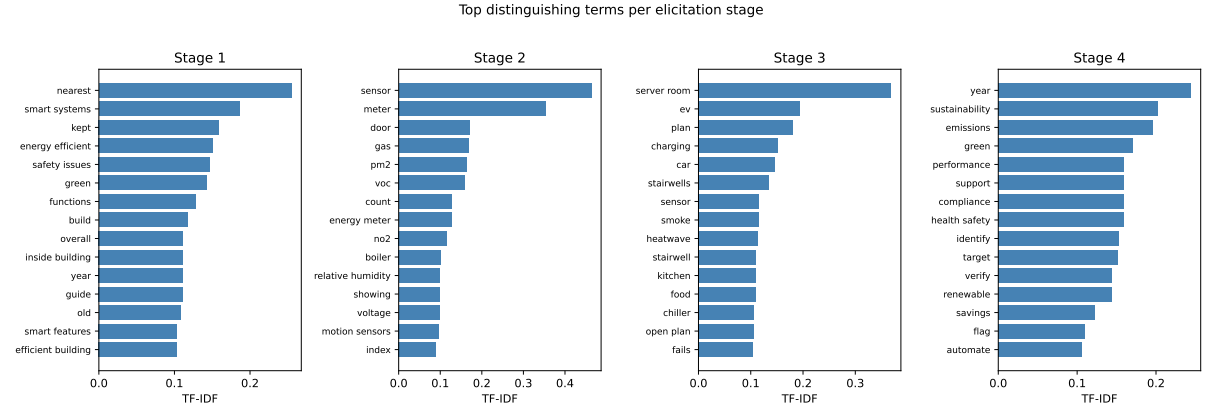


Fig. 17. The fifteen most distinctive terms of each stage. We joined each stage's questions into one document and weighted terms by TF-IDF across the four documents. We kept English words and two-word phrases (stop words removed) that occur in two or three stages.

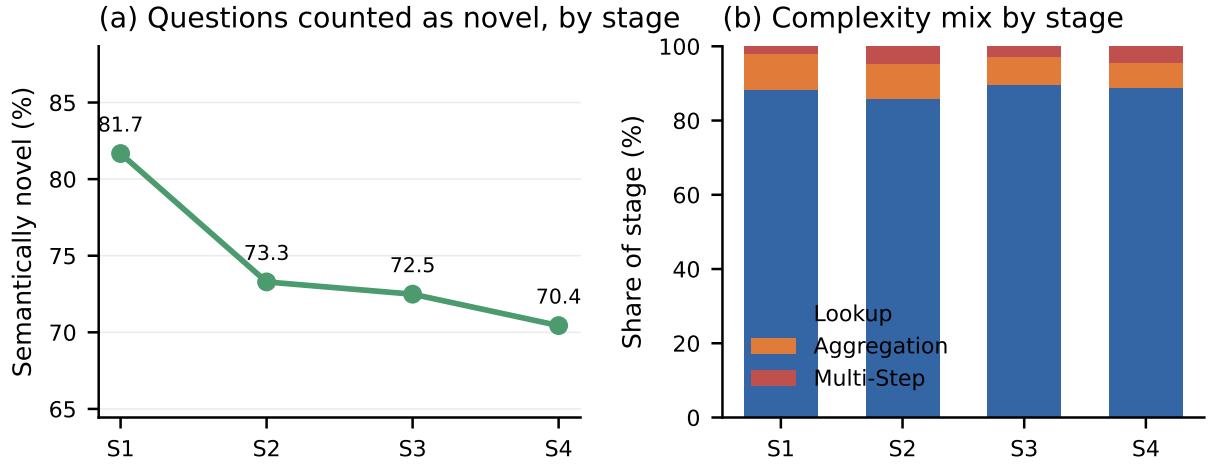


Fig. 18. (a) Share of each stage's questions counted as novel: TF-IDF cosine similarity below 0.75 to every earlier question in stage-then-time order. Much of the decline is built into the measure, which counts the first 500 questions as novel without comparison and compares later stages with more questions (§3.4). (b) Complexity labels by stage, from the automated classifier (agreement in Table 4).

(9.7%). The clearest persona concentrations are Health & Safety Officers in safety (63.1%) and Sustainability & Energy Teams in energy (56.0%). The persona figure omits the no-domain share, which is largest for Facility Managers (40.8%) and Building Owners (35.2%).

B.5 The topic ranking in full

Table 18 gives every topic's Borda score and mean rank over the 80 complete rankings by contributors. Table 19 gives Kendall's W within each persona and over all complete rankings (§3.2). Neighbouring topics in the top

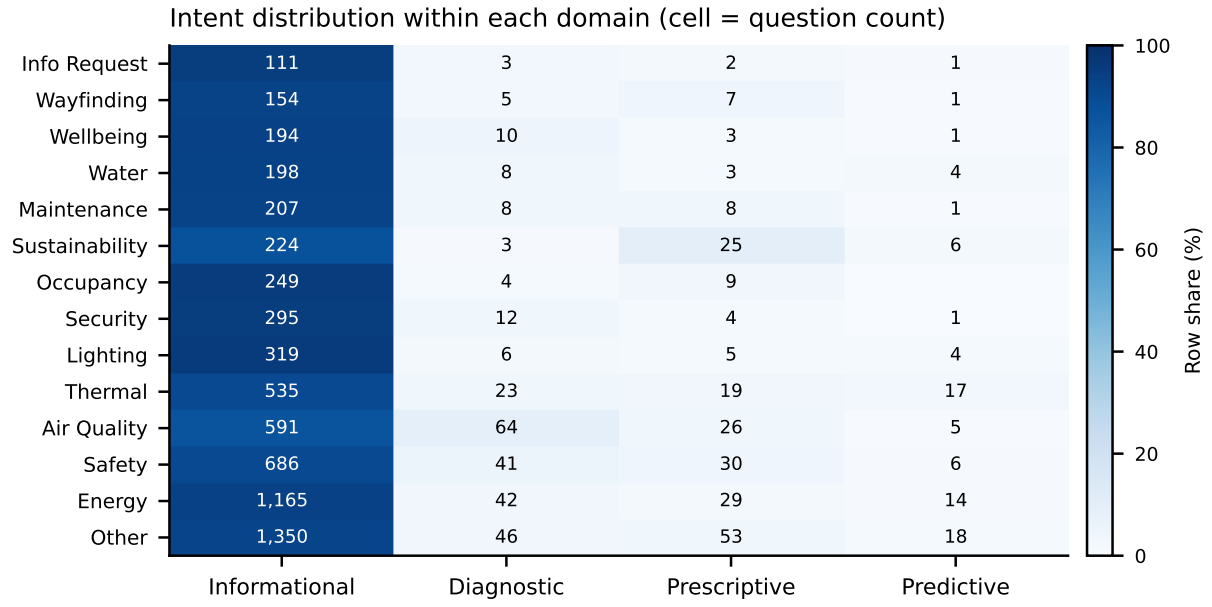


Fig. 19. Intent labels within each of the fourteen largest domains, from the automated classifier (agreement in Table 4). Cells give question counts; shading gives the share of the domain's questions.

ten lie at most 76 Borda points apart, and no single gap stands out, so the top of the ranking does not split into a leading group. Clustering the topics by how similarly contributors ranked them (average linkage on one minus the Spearman correlation between topics' ranks, cut into four groups) yields these groups: Energy Consumption, Water Management, Waste & Recycling, Renewable Energy & Solar, Green Spaces & Biodiversity and Carbon Footprint & Net Zero; Indoor Temperature Control, Air Quality & Ventilation, Lighting & Daylight, Noise & Acoustics, Security & Access Control, Occupancy & Space Utilisation, Fire Safety & Emergency, Health & Well-being and Building Maintenance & Faults; IoT Sensors & Data Analytics and Building Automation & AI; and Lifts, Stairs & Internal Transport, Parking & EV Charging and User Apps & Digital Interaction. These groups are not stable: the same procedure applied to every ranking account, including accounts that submitted no questions, groups the topics differently, so we draw no conclusions from them.

B.6 Depth preference by persona

Each contributor ranked the example questions of ten topics, so a persona's rankings in Table 20 come from one tenth as many people: 3 Building Owners, 4 Facility Managers, 5 IT Professionals, 8 members of Sustainability & Energy Teams, 10 Health & Safety Officers, 10 Occupants, 14 Students and 15 Guests. Figure 21 shows how contributors ranked the four example questions, overall and by persona, and Figure 22 plots each topic's priority against its Expert-first share (§3.3).

B.7 How hard questions are to ask and to answer, as rated by a language model

A language model (`gpt-5.5`, one run, at temperature 1, which the model requires) rated the 6,117 unique question texts, some more than once, on two six-level scales from one prompt. The *surface* level rates how hard a question is for a person to reason about, from factual recall to strategic reasoning. The *latent* level rates how hard it is

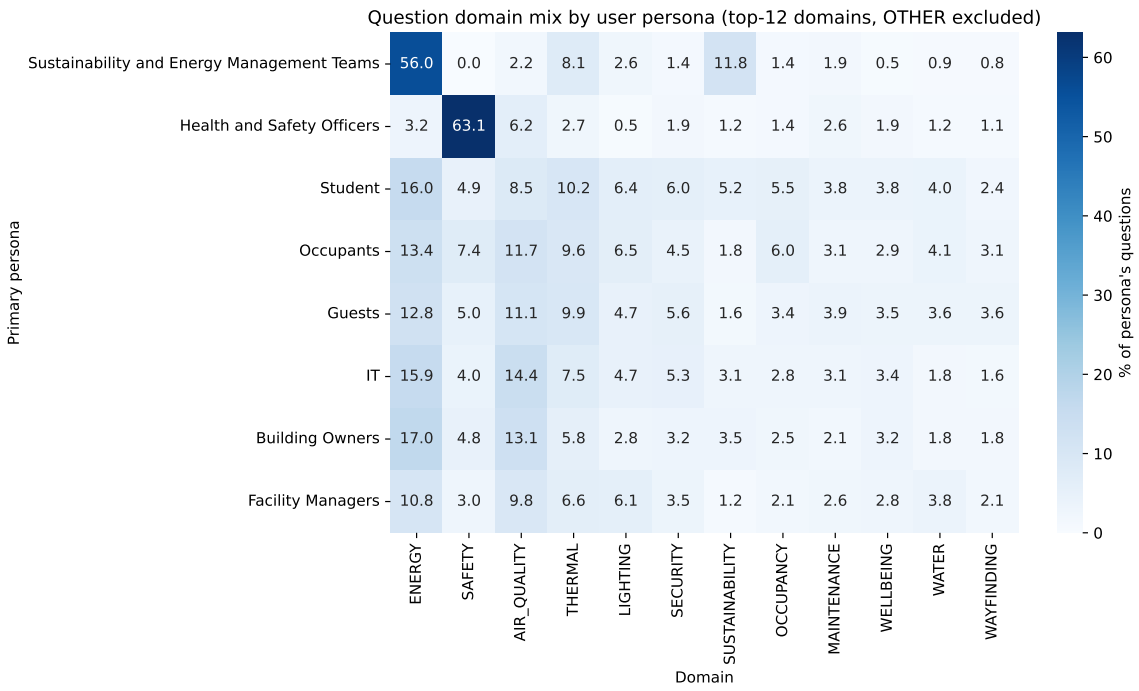


Fig. 20. Share of each persona's questions in each of the twelve largest domains other than no domain, from the automated classifier (agreement in Table 4). Each share is of all the persona's questions.

to answer with an automated building pipeline, judged against a list of the components of an earlier OntoSage build, not of the OntoSage+ studied in §5. The model also judged whether those components could answer the question fully, partly or only with an extension. The prompt told the model that every question can be answered by some combination of components. The ratings cover 6,157 rows, of which 40 repeat one of those texts, and no person checked them.

Figure 23 cross-tabulates the two levels. For 63.0% of rows, the model rated answering as harder than asking. It judged 22.9% of rows answerable in full with those components, 3.4% in part and 73.8% only with an extension. These ratings record what one model concluded about an earlier build; they measure neither the questions nor the system studied here. We report what answering the questions needs in §6, from the Part A codes.

B.8 Query types by volume and analytical share

Table 21 counts the corpus by query type and gives each type's analytical share: the share of its questions labelled `AGGREGATION` or `MULTI_STEP`. The shares of 100% follow from the classifier's rules (§B.1): a comparison, anomaly or historical question receives one of these labels either from a complexity keyword that matched first or, failing that, from its query type. The analysis script assigned the tier in the last column by a fixed rule: P1 when a type has at least 1,000 questions or more than 40% of them labelled analytical; P2 when it has at least 200 questions; and P3 otherwise. Topic rankings play no part in the rule. The three types analytical by rule hold 5.6% of the corpus, and the two largest types, in which the classifier labelled few questions analytical, hold 90.9%. Table 21 records what the analysis produced while we designed OntoSage+; the study did not test it.

Table 18. Borda score and mean rank of every topic over the 80 complete rankings by contributors (20 points for a first place, down to 1 for a last; a lower mean rank is a higher priority).

Rank	Topic	Borda	Mean rank
1	Air Quality & Ventilation	1,243	5.46
2	Indoor Temperature Control	1,207	5.91
3	Fire Safety & Emergency	1,137	6.79
4	Security & Access Control	1,107	7.16
5	Energy Consumption	1,036	8.05
6	Lighting & Daylight	960	9.00
7	Health & Well-being	952	9.10
8	Occupancy & Space Utilisation	886	9.93
9	Water Management	873	10.09
10	Noise & Acoustics	824	10.70
11	Waste & Recycling	815	10.81
12	Renewable Energy & Solar	797	11.04
13	Green Spaces & Biodiversity	722	11.98
14	IoT Sensors & Data Analytics	715	12.06
15	Building Maintenance & Faults	713	12.09
16	Lifts, Stairs & Internal Transport	601	13.49
17	Carbon Footprint & Net Zero	597	13.54
18	Building Automation & AI	560	14.00
19	Parking & EV Charging	556	14.05
20	User Apps & Digital Interaction	499	14.76

Table 19. Agreement among contributors on the ranking of twenty topics (Kendall's W), within each persona and over all complete rankings. Smaller groups show higher W by chance (§3.2).

Persona	Rankers	Kendall's W
Sustainability & Energy Teams	9	0.614
Health & Safety Officers	10	0.533
Occupants	13	0.508
Students	14	0.415
Building Owners	5	0.379
Guests	17	0.325
Facility Managers	5	0.306
IT Professionals	7	0.305
All contributors	80	0.227

Table 20. Share of each persona's rankings that placed each level first (the instrument's Basic, Intermediate, Advanced and Expert; §3.3), by contributors only. *n*: the persona's rankings, ten per contributor. Bold: the persona's largest share.

Persona	*n*	Basic	Intermediate	Advanced	Expert
Occupants	100	**58.0**	18.0	6.0	18.0
Students	140	**47.9**	24.3	14.3	13.6
Building Owners	30	**43.3**	26.7	20.0	10.0
Guests	150	**36.0**	22.0	13.3	28.7
IT Professionals	50	18.0	**44.0**	30.0	8.0
Facility Managers	40	17.5	35.0	**37.5**	10.0
Health & Safety Officers	100	15.0	14.0	17.0	**54.0**
Sustainability & Energy Teams	80	8.8	8.8	16.3	**66.3**

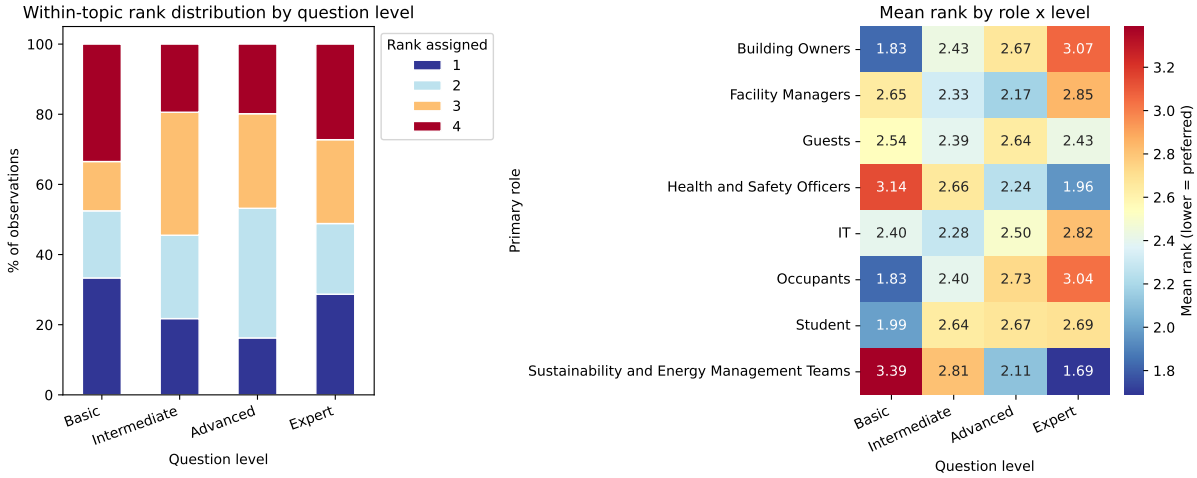


Fig. 21. Preferences among the four example questions per topic, from the 690 rankings by contributors. Left: share of rankings placing each level first, second, third or fourth. Right: mean rank of each level by persona (lower is preferred).

Table 21. Query types in the corpus by volume and by the share the automated classifier labelled `AGGREGATION` or `MULTI_STEP` (agreement in Table 4), with the tier the analysis script assigned by a fixed rule (§B.8).

Query type	Questions	%	Analytical %	Tier
STATUS	4,793	67.0	8.5	P1
CAPABILITY	1,704	23.8	1.5	P1
COMPARISON	175	2.5	100.0	P1
ANOMALY	160	2.2	100.0	P1
HISTORICAL	68	1.0	100.0	P1
RECOMMENDATION	128	1.8	12.5	P3
DIAGNOSTIC	123	1.7	4.9	P3

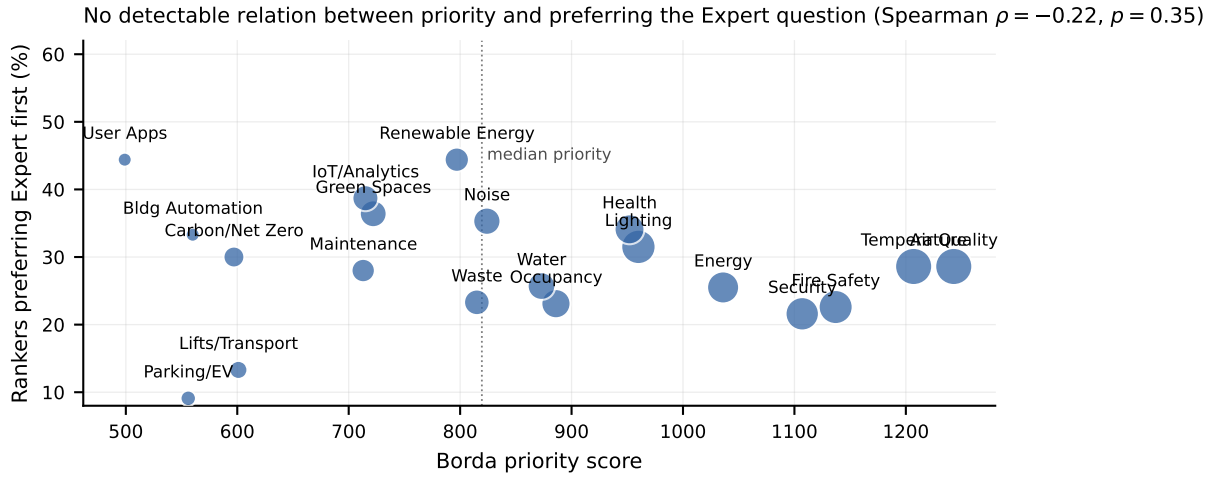


Fig. 22. Topic priority against wanted depth, over contributors only: each topic's Borda score and the proportion of contributors ranking its example questions who put the Expert question first. Point size grows with the number of those contributors. The two showed no detectable relation (Spearman $\rho = -0.22$, $p = 0.35$).

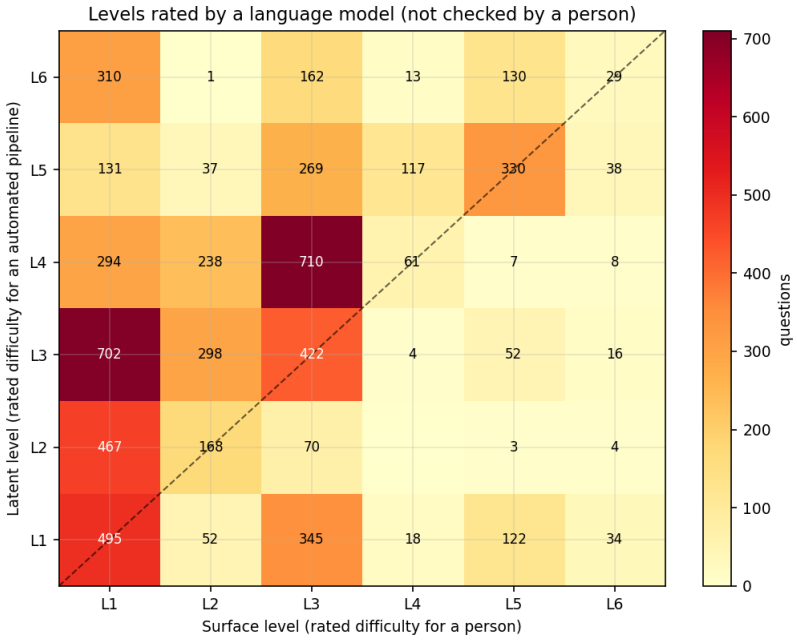


Fig. 23. A language model's ratings of the questions on two six-level scales: how hard each is for a person to reason about (surface level, horizontal) and to answer with an automated building pipeline (latent level, vertical). Cells count rated rows; cells above the diagonal are rows rated harder to answer than to ask. No person checked the ratings, and the model judged the latent level against an earlier OntoSage build.

C Count Reconciliation

Table 23 recomputes each count used in §3 from the survey's exported files.

Ranking accounts that are not contributors. Stage 5 saved each ranking under the account that made it. Of the 91 accounts that ranked all twenty topics, 80 match a contributor's username and 11 do not: 3 carry a research-team member's name, 4 more were created before the first corpus question was submitted and the other 4 were created during the survey. Of the 76 accounts that ranked example questions, 69 are contributors and 7 are not; all 7 unmatched accounts also ranked the topics. The exports do not reveal whether an unmatched account was a test account, a person using a different name or one whose questions were removed in screening. We therefore exclude all unmatched accounts. Every ranking result in §3 and Appendix B uses the 80 topic rankings and the 690 question rankings made by contributors. The original analysis scripts computed these results on copies of the ranking files without the unmatched accounts. Table 22 shows what the exclusion changes: 2 of the twenty topics change position (0 in the top ten), and each result keeps its interpretation.

Entries that are not questions. The research team screened submissions per contributor and kept every entry of an accepted contributor. The corpus therefore still holds 63 single-word entries, such as greetings, which the classifier labels as matching no domain (§B.1).

D Departures from the Study Plan

The study plan was approved on 13 October 2026, before the first session on 19 October; we did not register it publicly. Tables 24 and 25 list every departure, each with what we planned, what we ran and the consequence for the analysis.

E Agreement Between the Coders in Full

The second coder (the third author) labelled seeded samples of five materials without seeing the AI's proposals or the first author's labels (§5.5); the Stage 5 answers had no second coder. Tables 26 and 27 give the agreement on every field, the counts behind each field with agreement below 0.60 and how the disagreements were settled.

Table 22. Stage 5 ranking results as reported, over contributors only, and over every ranking account, including the accounts that submitted no questions.

Result	Contributors only (reported)	All ranking accounts
Complete topic rankings	80	91
Kendall's W over the topic rankings	0.227	0.206
Building Automation & AI, position of twenty	18	19
Parking & EV Charging, position of twenty	19	18
Within-topic rankings	690	760
Basic placed first (%)	33.3	34.9
Expert placed first (%)	28.7	27.5
Basic placed last (%)	33.5	31.5
Expert placed last (%)	27.3	29.0
Lowest mean rank	2.477	2.428
Highest mean rank	2.522	2.543
Spearman ρ, Borda against Expert-first share	−0.22	−0.25
p of that ρ	0.35	0.29

Topics in a different position in the two rankings: 2 of twenty (Parking & EV Charging; Building Automation & AI), 0 of them in the top ten.

Table 23. Counts in the survey's exported files, each recomputed from the exports by a script.

Quantity	Count	How it was counted
Contributors (Stages 1–4)	96	Distinct usernames in the question export. Each chose one or more of fourteen role profiles: 80 chose one and 16 chose several.
Question submissions	7,151	Rows in the question export: 1,904 / 2,003 / 1,705 / 1,539 in Stages 1–4.
Unique question texts	6,117	After folding case, spacing and end punctuation (1,034 repeats); 6,117 rows in the de-duplicated export. 464 texts occur more than once, 431 of them from two or more contributors.
Single-word entries kept	63	Submissions were screened per contributor, not per entry.
Stage 5 open-prompt respondents	59	295 answers to five prompts, 9,396 words; all 59 are contributors.
Topic rankings (all 20 topics)	91	80 by contributors. 11 accounts match no contributor: 3 carry a research-team name, 4 more were created before the first corpus question, and 4 others. This article uses the 80 by contributors.
Within-topic depth rankings	760	From 76 accounts, 10 topics each. 69 accounts are contributors; 7 are not (70 rankings, 2 of these accounts with a research-team name). This article uses the 690 by contributors.

Accounts are matched to contributors by username. An account that matches no contributor may be a person whose questions were removed in screening, a person who used a different name, or a test account; the exports cannot tell these apart.

On the live answers, the coders differed most at the boundary between W and WL: of the 30 answers, the first author labelled 4 as W and 11 as WL; the second coder, 10 and 2. Table 28 gives every pair of labels. On the reasons, agreement on two codes the results use fell below 0.60 because the second coder applied them far less often: whether a named limit was the key's limit (11 times by the first author, 1 by the second coder) and whether the participant took a refusal as the answer (15 and 3 times). On the Part A and the mixed questions, agreement on the place and criterion fields fell below 0.60; in both samples, the second coder coded a place or criterion as missing more often than the first author. The second coder's place, time and criterion codes come from a re-code under the revised codebook, which added a not-applicable rule for place and time (Appendix D); we do not use the first pass.

Table 29 gives each result that rests on the live labels or the two reason codes in two versions: as reported, and with the second coder's label or codes replacing the agreed ones for each item the second coder coded. The correct-decision rates by answer received keep their order: highest after a supported answer and lowest after a wrong refusal. With the second coder's codes, fewer named limits match the key and fewer reasons take a refusal as the answer. Most named limits in S2 and S3 still match the key and none in S1 does. More than half of the reasons still take the refusal as the answer, both after a wrong refusal and after a correct refusal or a stated limit. Because only the sampled items change, the second version shows how far the second coder's reading moves each result on the items the second coder saw, not what that reading would give on every item.

Table 24. Departures from the study plan.

Item	Planned	As run	Consequence
Public registration	file the analysis plan publicly before the first participant	not filed; the plan, keys and rubric were written and approved before the first session (13–14 Oct; first session 19 Oct)	the order of plan and data rests on dated project files, not a public record
Pilot	three pilot sessions	not run (the supervising author's decision); one person new to OntoSage+ did a full self-guided test run instead	exclusion rule 4 (time below the pilot median x 0.3) has no threshold; reported as not applicable
Session mode	as in the approved protocol	online and self-guided, through a public tunnel; the ethics committee confirmed the changes	none on the measures; mode recorded per participant
Study form after P01	decide after P01: add a safeguard or keep the form	kept unchanged: participants could ask with the decision options visible	exploratory split by where the first question was asked (RQ3)
Configuration	frozen 14 Oct (hash 4ccb69bc...)	STATIC_BASE_URL added 15 Oct so images load online (hash 204d8062...); orchestrator recreated 20 Oct after an unclean shutdown	answers' content unaffected; logged in the runbook
Linking questions to tasks	by task (a new chat for each task)	by the page on screen when the question was sent (each goal page asked for a new chat, but most participants kept one chat across tasks)	answer received = last answer before the decision; a worst-answer sensitivity is reported (RQ3); a room or date can carry from one task into the next (Limitations)
E1 model	glmer code + form	singular; declared fallback (code only) also singular	bootstrap and GEE beside it; exploratory refit without S3 shows the form effect
Answer labels	the first author labels every answer; a second coder labels seeded samples; disagreements settled	an AI model proposed every label; the first author reviewed each proposal and kept or changed it; the second coder labelled the seeded samples without seeing either; the A-minus proposals were made as the run produced them, so not blind to the system	the first author's labels started from the AI's proposals; agreement is between the second coder and the first author's reviewed labels
Settling disagreements	the coders discuss each one; if they still disagree, the supervisor decides	the supervisor was the second coder, so on 25 Oct, before any disagreement was seen, the first author and the second coder agreed a default: a label or code still disputed keeps the first author's. For the live answers, every disagreement (12) was instead discussed individually against the rubric on 28 Oct, after both sets of labels were in – 9 confirmed the first author, 3 the second coder; the other four materials fell back to the default, confirmed by the second coder in discussion but not reviewed item by item	the number still on the default is reported, with those results on the second coder's labels as well; the live-answer discussion and its rationale are kept with the data

Table 25. Departures from the study plan (continued).

Item	Planned	As run	Consequence
Second-coder samples	drawn after the first author's labels	drawn on 24 Oct, before the first author's review; the first author did not open them until the review was finished	the first author could not know which items were double-coded
Part A codebook	refined on 100 development rows, then frozen	no development session; frozen as drafted (24 Oct)	the definitions were not tested on development rows before coding
Part A codebook, F3a/F3b 'n/a'	one frozen definition, the same for both coders	v1 (24 Oct) defined 'n/a' only for F3c; in the second coder's first pass no F3a or F3b code was 'n/a' on any of the 160 sampled items; v2 (28 Oct) added the missing rule, and the second coder re-coded F3a/F3b/F3c on those items under it, without the first author's codes	every F3a/F3b/F3c agreement value reported (Part A and mixed questions) comes from the re-code under v2; the first pass is not used; F3c's own definition is unchanged in v2
Comparison assistant	a JSON action loop through the shared model client	the provider's native tool calls, same model, endpoint, temperature and token budget (recorded 10 Oct)	a model trained to call tools is the fairer baseline
Offline runs	as specified	two runners shared the stack; two non-study accounts; the assistant's step-limit non-answers kept as its answers; 8 gateway failures and 1 failure of our run script (the baseline launcher did not start) re-run once; 2 failed again and are scored PF	timing comparable within the run only
Two-checks-off run (A-minus)	after the last session; settings restored and checked	run 24 Oct after the last session; restored with every check passing (configuration hash as logged); its container's own log was not kept, so its check records come from each answer's route	how often the checks acted is counted for OntoSage+ only; with the checks off they cannot act
Qualitative and question coding	the first author codes everything; a second coder codes seeded samples	an AI model proposed codes for every question and reason with the codebooks; the first author reviewed each proposal and kept or changed it; the second coder coded seeded samples without seeing either; disagreements settled	the first author's codes started from the AI's proposals; agreement is between the second coder and the first author's reviewed codes
Stage 5 codes	one coder, re-coding a seeded 10% a week later	an AI model proposed codes with the codebook; the first author reviewed each proposal and kept or changed it; no re-coding a week later	the codes started from the AI's proposals; one reviewer, and no measure of how consistently the codes were applied

Table 26. Agreement between the two coders for every field, and how the disagreements were settled: the answers and the reasons.

Material and field	n	κ [95% CI]	Same	First author	Second coder
Answers participants received (RQ3)					
answer label	30	0.48 [0.28, 0.69]	60%	W 4, WL 11, FD 11, U 2, OT 2	W 10, WL 2, FD 12, U 2, OT 4
U against every other label	30	1.00			
labelled differently, each discussed against the rubric	12			kept: 9	taken: 3
Offline answers (RQ2)					
answer label	130	0.93 [0.87, 0.97]	95%		
U against every other label	130	1.00			
labelled differently, settled by the rule	8			kept: 8	
Reasons participants wrote (RQ3)					
all codes: median (lowest to highest)	36	0.83 (0.00 to 1.00)	93% of cells		
named a limit of the answer	36	0.83			
the limit named is the key's limit	36	0.12	72%	ticked 11	ticked 1
took a refusal as the answer	36	0.23	67%	ticked 15	ticked 3
cited what the answer showed	36	0.94			
would check elsewhere	36	1.00			
asked again in other words	36	1.00			
trusted detail or a confident tone	36	0.00	97%	ticked 1	ticked 0
frustrated by a refusal	36	0.00	97%	ticked 1	ticked 0
accepted a related reading	36	0.00	97%	ticked 1	ticked 0
went on prior knowledge	36	not defined	100%	ticked 0	ticked 0
guessed	36	not defined	100%	ticked 0	ticked 0
other	36	1.00			
reasons coded differently, settled by the rule	20			kept: 20	

Agreement between the second coder and the first author's reviewed labels and codes, on the seeded samples the second coder labelled without seeing any other label. κ: Cohen's kappa; a 95% interval (bootstrap) was computed for the two answer-label sets only. Same: the share of items (for the reasons, of code cells) given the same value. (J): mean Jaccard similarity of the two coders' sets of codes, for the two fields that take several codes. First author and second coder: how often each used the value, given for every field below 0.60; for answer labels, the number of answers given each label; for reason codes, the number of the reasons each coder ticked; for place, time and criterion, the number coded specified / missing / n/a. A κ of 0.00 with most items the same arises when one coder used a code once and the other never. In the rows on settling, n is the number of items the coders labelled or coded differently; kept: those that kept the first author's label or code; taken: those settled in the second coder's favour. For the live answers every disagreement was discussed one by one against the rubric; for the other materials a disagreement kept the first author's label or code by a rule agreed before either set was compared (Appendix D).

Table 27. Agreement between the two coders for every field, and how the disagreements were settled: the questions, the Stage 5 answers, and all materials together.

Material and field	n	κ [95% CI]	Same	First author	Second coder
Survey questions, Part A (RQ1)					
operation asked for (mean Jaccard)	100	0.88 (J)			
what a complete answer needs (mean Jaccard)	100	0.70 (J)			
names the place	100	0.34	58%	40 / 33 / 27	52 / 47 / 1
gives the time	100	0.75	84%		
defines the criterion	100	0.54	73%	9 / 23 / 68	19 / 35 / 46
could the building's data establish it	100	1.00			
answerable in principle	100	0.67			
states a purpose	100	1.00			
questions coded differently, settled by the rule	83			kept: 83	
Participants' own and corpus questions, mixed (RQ1)					
operation asked for (mean Jaccard)	60	0.86 (J)			
what a complete answer needs (mean Jaccard)	60	0.78 (J)			
names the place	60	0.24	47%	25 / 6 / 29	27 / 25 / 8
gives the time	60	0.76	88%		
defines the criterion	60	0.45	78%	4 / 8 / 48	5 / 12 / 43
could the building's data establish it	60	1.00			
answerable in principle	60	0.80			
states a purpose	60	1.00			
questions coded differently, settled by the rule	50			kept: 50	
Stage 5 answers (RQ1)					
every code		no second coder			
All materials					
items labelled or coded differently	173			kept by the rule: 161	

Notes as for Table 26.

Table 28. The live answers both coders labelled: the first author's label against the second coder's.

First author	W	WL	FD	U	OT	Total
W	3	1	0	0	0	4
WL	6	1	2	0	2	11
FD	1	0	10	0	0	11
U	0	0	0	2	0	2
OT	0	0	0	0	2	2
Total	10	2	12	2	4	30

Rows: the first author's label; columns: the second coder's, before the disagreements were settled. The coders gave the same label to 18 of the 30 answers (the diagonal). Of the 12 they labelled differently, 7 lie between W and WL.

Table 29. The results that rest on the fields with the lowest agreement, as reported and with the second coder's labels or codes.

Result	As reported	Second coder's labels or codes
Correct decisions by the last answer received (live labels, $\kappa = 0.48$)		
after W (supported by the data)	28/35 (80%)	78.0% of 41
after WL (states its limits)	34/47 (72%)	75.0% of 40
after FD (wrong refusal)	5/42 (12%)	11.6% of 43
after U (not supported by the data)	16/32 (50%)	50.0% of 32
after OT (answers something else)	4/9	44.4% of 9
no answer before deciding	8/15 (53%)	53.3% of 15
Reasons that named a limit ($\kappa = 0.83$) and whether it was the key's limit ($\kappa = 0.12$)		
reasons that named a limit	105	106
of which the key's limit	67	57
S1: the key's limit, of those naming a limit	0 of 15	0 of 15
S2: the key's limit, of those naming a limit	35 of 37	30 of 36
S3: the key's limit, of those naming a limit	18 of 18	14 of 18
S4: the key's limit, of those naming a limit	14 of 35	13 of 37
Reasons that took a refusal as the answer ($\kappa = 0.23$)		
after a wrong refusal (FD)	31 of 43	27 of 43
after a correct refusal or a stated limit (WL)	30 of 46	25 of 46

As reported: the agreed labels and codes behind the results in §8. Second coder's labels: the same computation with the second coder's label or codes put in place of the agreed one for each item the second coder coded (30 live answers, 36 reasons); every other item keeps its agreed label or code. Percentages of decisions that were correct.

F Further Study Tables and Figures

The body cites these tables and figures, but readers can follow it without them.

The building's model. Table 30 counts Building A as modelled in the graph frozen for the study (§4.3).

Records. Table 31 counts the records the study produced, from the participants who completed a session to the offline runs that reused their questions.

Decisions on each form. Table 32 and Figure 24 give the number of correct decisions and the options participants chose on each of the twelve forms (§8). Forms within a situation differed widely: in S1, correct decisions ranged from 7/10 on S1-1 to none on S1-2 and S1-3, and in S4 from 17/20 on S4-2 to 3/20 on S4-3. This spread is the variation between forms that the mixed model found.

Decisions, ratings and measures. Table 33 gives the correct decisions by situation that Figure 10 plots, and Figure 25 shows the decisions by the label of the last answer before deciding, which Table 13 counts (§8). Figure 26 shows how participants rated the answers to their own questions asked live, Figure 27 each participant's SUS, trust and mental-effort scores (Table 14) and Figure 28 how participants worked.

Participants' own questions against corpus questions. Table 34 gives the shares behind Figure 7 with their counts (§6) and adds whether each question is answerable in principle.

The evidence panel against the label. Table 35 cross-tabulates the kind of statement OntoSage+'s evidence panel named against the label of the answer it accompanied (§7).

Table 30. Building A as modelled in the graph frozen for the study, counted by SPARQL queries against it. Each point is counted once, under the first kind whose classes it carries, so the kinds sum to the total. Brick classes a rooftop and a parking level as kinds of floor, so the floor nodes are the building's 6 storeys and these 2 spaces. Besides the stores that points name, the store registry lists a store of interval records (bookings, work orders, door access and equipment outages), which no point names.

Characteristic	Count
Spaces and equipment	
Floor nodes (`brick:Floor` and its subclasses)	8
Locations (`brick:Location` and its subclasses)	298
Equipment (`brick:Equipment` and its subclasses)	147
Points, by kind (each counted once)	
Indoor climate (CO_2, humidity, air temperature)	863
Air quality (particulates, TVOC, NO_2, CO, formaldehyde, gas)	765
Lighting (colour temperature, illuminance)	525
Occupancy (count, status)	500
Door and window contacts, intrusion detectors, cameras	481
Sound and noise level	234
HVAC plant (air flow, damper, filter, fan)	36
Waste (fill, weight)	24
Electrical power and energy	22
Water (flow, level, temperature, metering)	21
Outdoor and weather	4
Not assigned to a kind above	37
Points in all	3,512
Distinct point classes	86
Links from points to stored readings	
Timeseries references (`ref:hasTimeseriesId`)	3,572
distinct series identifiers	3,545
Stores named by points (distinct `ref:storedAt` keys)	21
Extensions to Brick	
Lay-term concepts (HBCO)	103
lay terms	728
Amenities (OCBV)	75
Knowledge topics (OCBV)	50
Classes declared in the OCBV namespace	107
Properties declared in the OCBV namespace	340

The offline comparison by question set. Figure 29 repeats Figure 8 for each question set, Figure 30 gives the labels per system and set, and Table 36 counts the questions whose label changed between three runs of the same question (§7).

Time to answer offline. Table 37 gives each system's time per answer in the offline comparison and how often Assistant B stopped at its step limit without answering (§7).

The canonical task questions. Figure 31 shows each system's labels for the three runs of each form's canonical question, its question in the researchers' wording. OntoSage+'s answers were not supported by the data in 30.6% of these runs and Assistant B's in 8.3%; OntoSage+'s answers to S2-1, S4-1 and S4-3 were labelled unsupported in all three runs. These labels have no second coder.

Table 31. Records from the study, from participants to the offline runs.

Record	n
Participants who consented and completed the session	30
Decisions (six tasks each)	180
Questions typed to OntoSage+	391
while a task was on screen	274
own questions asked live	114
practice or other pages	3
Own questions written before using OntoSage+	85
Tasks decided with no question asked	4
Offline comparison answers (RQ2), rows	203
Canonical-question runs (12 × 3 × 2), rows	74
Stability runs (30 × 3 × 2), rows	184

Counts reconcile with the session store, Open WebUI and OntoSage+'s turn memory (reconciliation.json). Matched to turn memory: 370.

Table 32. Decisions on each of the twelve study forms.

Form	Correct	Correct option(s)	Options chosen	Median min	Mean confidence
S1-1	7/10	a	a:7 d:3	4.0	84
S1-2	0/10	a	c:1 d:9	3.6	85
S1-3	0/10	b	c:5 d:5	2.7	78
S2-1	14/20	b	a:6 b:14	3.3	86
S2-2	0/20	c	a:3 e:17	2.8	66
S2-3	12/20	c	a:1 c:12 d:3 e:4	2.4	87
S3-1	10/10	b d	b:7 d:3	3.2	90
S3-2	10/10	b d	b:7 d:3	2.4	87
S3-3	10/10	b d	b:9 d:1	2.1	84
S4-1	12/20	d	b:2 d:12 e:2 f:4	4.1	91
S4-2	17/20	b	a:2 b:17 d:1	2.2	84
S4-3	3/20	a	a:3 b:15 d:2	3.1	88

A decision is correct when it is the option the building's data support (fixed in the answer key before the study).

Table 33. Correct decisions by situation.

Situation	Correct	%	95% CI (bootstrap)	95% CI (GEE)
S1 data hold the answer	7/30	23	[10, 40]	[12, 41]
S2 data hold part of it	26/60	43	[32, 53]	[33, 55]
S3 data do not hold it	30/30	100	[100, 100]	not estimable
S4 two facts to combine	32/60	53	[42, 65]	[41, 65]
All	95/180	53	[48, 57]	

A decision is correct when it is the option the building's data support (fixed in the answer key before the study). Participant-cluster bootstrap, 2,000 resamples. In S3 both 'not recorded' and 'could not find out' are correct: of 30 S3 decisions, 23 said the data do not hold it, 7 chose 'could not find out', 0 accepted a value. Every S3 decision was correct (30/30), so the model cannot give an interval for it.

Table 34. Participants' own questions against corpus questions (Part A codes).

Code	Own (n = 85)	Corpus (n = 90)	Own minus corpus, points [95% CI]
What a complete answer needs			
sensor readings	36% (31)	34% (31)	+2 [-15, +20]
the building's description	54% (46)	26% (23)	+29 [+12, +44]
dated records	16% (14)	10% (9)	+6 [-4, +18]
policies and manuals	22% (19)	37% (33)	-14 [-30, +1]
outside knowledge	4% (3)	24% (22)	-21 [-34, -9]
no evidence, or cannot tell	0% (0)	7% (6)	-7 [-12, -2]
Could the building's data establish it?			
yes, directly	91% (77)	64% (58)	+26 [+13, +39]
only through a related quantity	9% (8)	12% (11)	-3 [-13, +8]
no building data could	0% (0)	18% (16)	-18 [-28, -8]
cannot tell	0% (0)	6% (5)	-6 [-10, -1]
Answerable in principle?			
yes	84% (71)	58% (52)	+26 [+10, +42]
partly	16% (14)	22% (20)	-6 [-18, +8]
no	0% (0)	20% (18)	-20 [-31, -10]

Own questions: written by the 30 participants before they used OntoSage+. Corpus: 90 questions drawn on 9 Oct from the earlier survey corpus for this comparison. Coded blind to source with the Part A codebook (v1, frozen 24 October 2026); 'what the answer needs' can take more than one value, so its shares do not sum to 100%. Intervals resample contributors within each source (2,000 resamples). 6 items repeat another item's text within the same source (none across sources); without them no difference moves by more than 1.5 points.

Table 35. What the evidence panel said the answer was, against the label of the answer.

The panel said the answer was	W	WL	FD	U	OT	All
calculated	27	10	23	34	7	101
inferred	0	1	0	0	6	7
not assessable (evidence could not support an answer)	5	17	17	0	8	47
observed (read from an instrument)	37	61	87	15	11	211
predicted	0	0	0	0	1	1
All	69	89	127	49	33	367

Live answers that carried a panel: 367 of 391. Rows: what the first line of OntoSage+'s panel said the answer was. The panel does not follow what the answer did: 110 of 127 wrong refusals (FD) came with a panel saying the answer was observed or calculated, as did all 49 answers not supported by the data (U), and 5 of 118 answers labelled W or U came with 'not assessable'. A consistency check, not the rubric's judgement of whether the listed sources support the text (that needs the human pass: the panel names sources by kind, not by room or period). Labels: the AI's proposals as reviewed by the first author, with the disagreements on the second coder's sample settled.

Table 36. Questions whose label changed between three runs of the same question.

Question set	OntoSage+	Assistant B
Canonical task questions (12, three runs)	17%	25%
Participants' task questions (30, three runs)	17%	20%

A change is any difference in the label (W, WL, FD, U, OT) between the three runs of one question, all caches flushed before every answer. Labels: proposed by an AI model on the blind sheet and reviewed by the first author alone (no second coder).

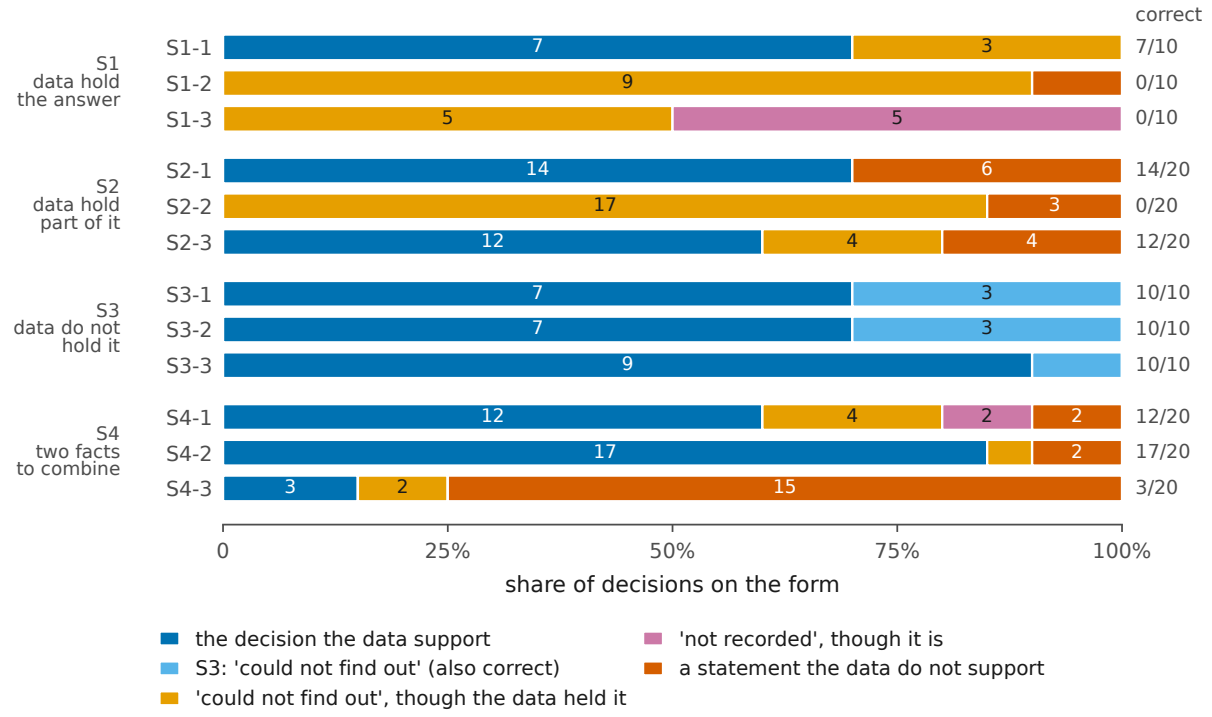


Fig. 24. What participants decided on each form: the decision the data support; “could not find out” or “not recorded” where the data held the answer; or a statement the data do not support. In S3, “could not find out” is also correct. The right-hand column counts correct decisions on each form.

Table 37. Time to answer in the offline comparison.

System	Answers	Median s	IQR s	90th pct s	Timeouts	Empty answers
OntoSage+	100	33	20-48	69	0	0
OntoSage+, two checks off	100	23	10-30	40	0	0
Assistant B	100	20	12-29	69	0	13

Seconds measured by the run script, caches flushed before every answer. OntoSage+ and Assistant B ran while two runners shared the stack; the two-checks-off run ran alone, so its times are not comparable with theirs (and it also skips the answer-relevance check’s model call). Timeouts: answers cut off at the run script’s 420-second limit. Empty answers: the assistant reached its step limit without answering (scored as a decline under the rubric).

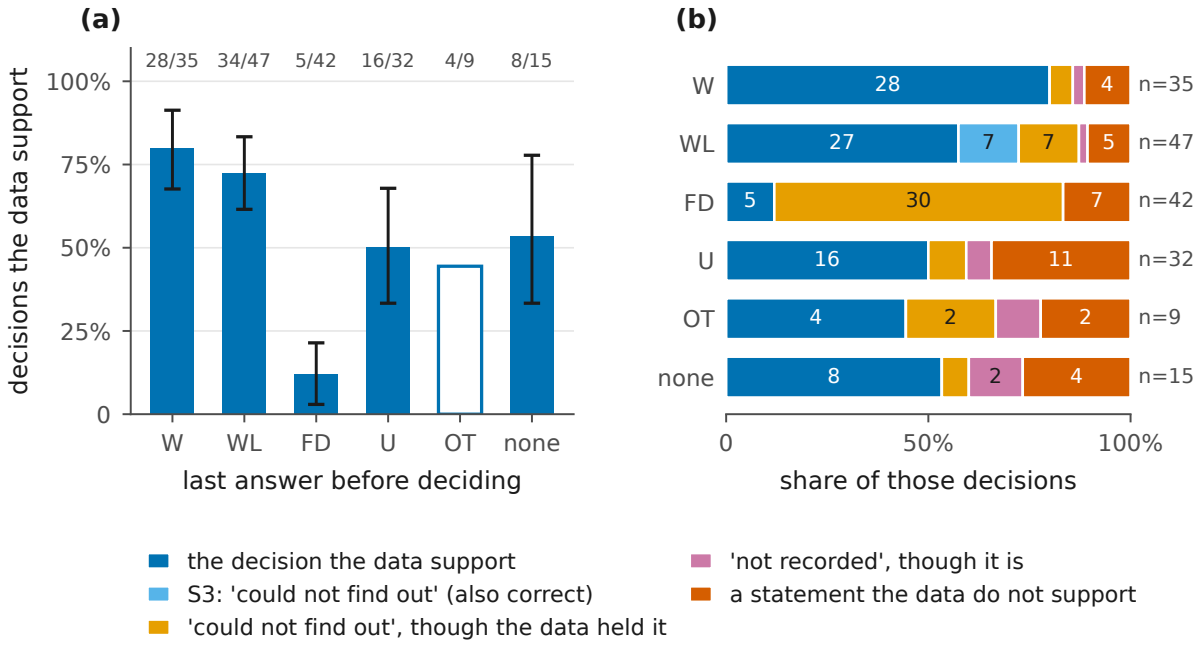


Fig. 25. Decisions by the label of the last answer before deciding: share correct (a) and what participants chose (b).

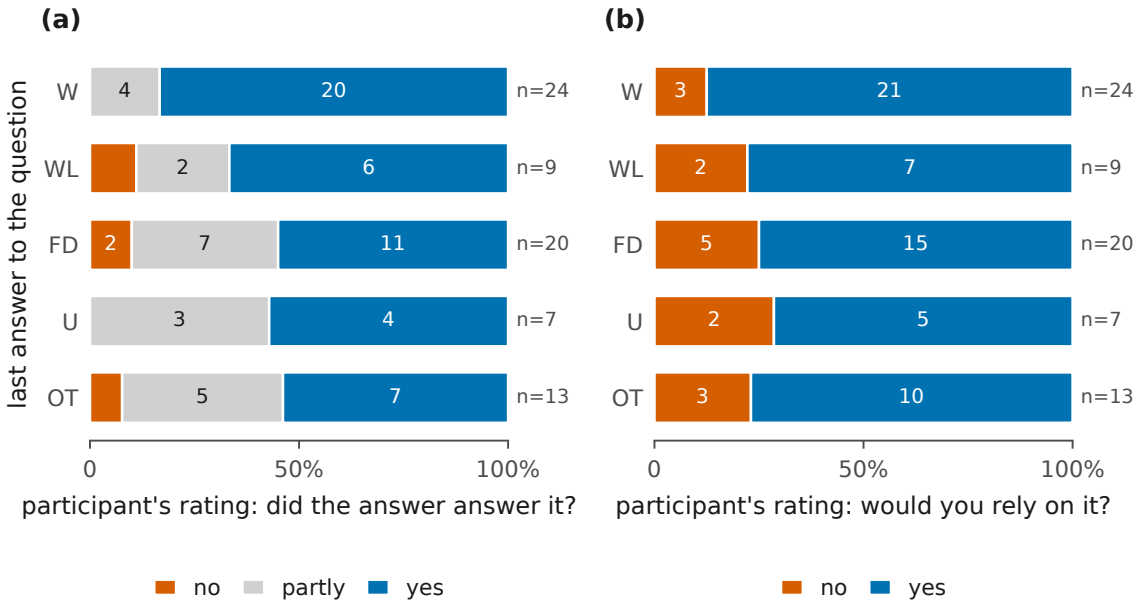


Fig. 26. Participants' own questions asked live: how they rated the answer, by the label of the question's last answer; n is the number of questions.

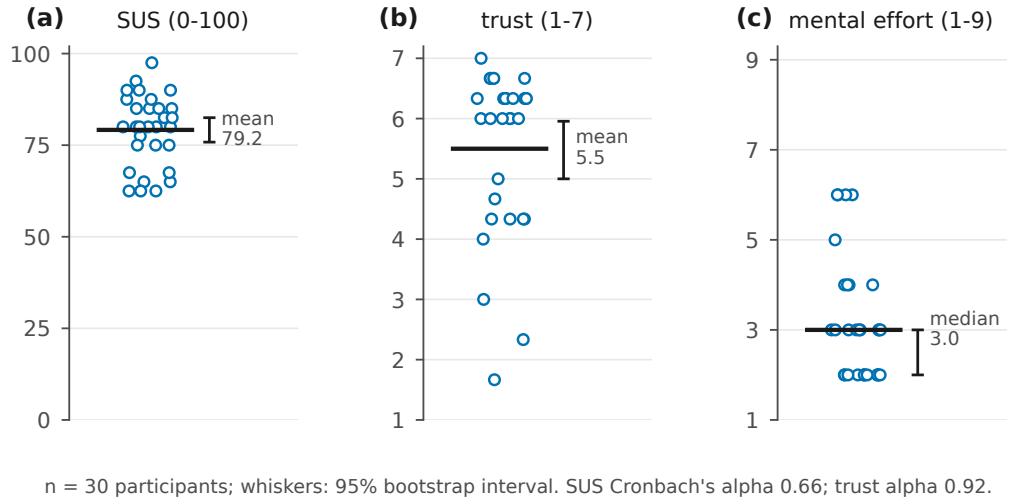


Fig. 27. SUS, trust and mental effort, one dot per participant.

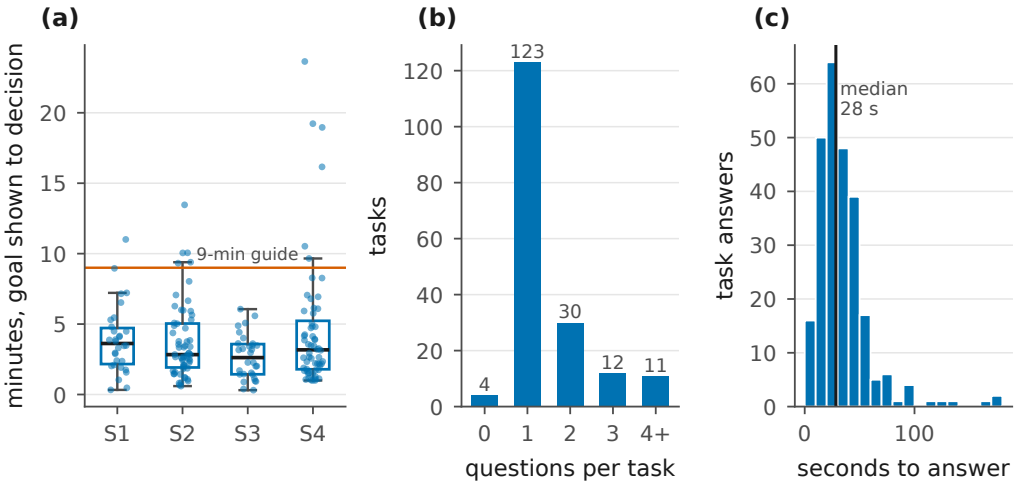


Fig. 28. How participants worked: minutes from the display of the goal to the decision, by situation, with a line at nine minutes (a); questions per task (b); and the seconds OntoSage+ took to answer questions asked during tasks (c).

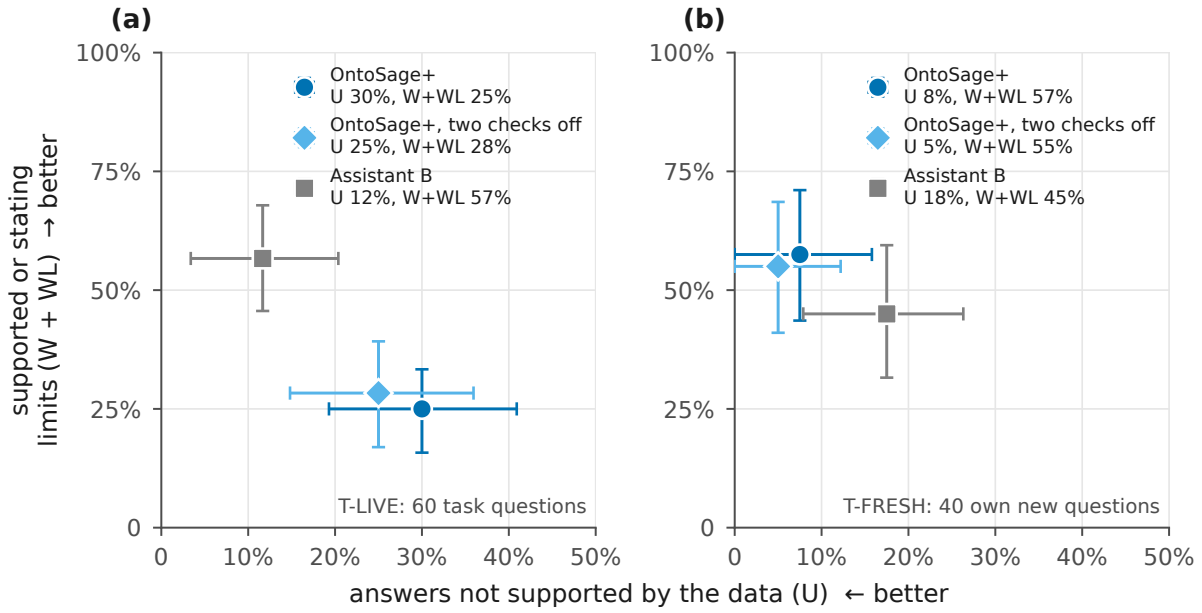


Fig. 29. As Figure 8, by question set: (a) T-LIVE, the first questions participants typed in a task; (b) T-FRESH, their own questions, written before using OntoSage+ and new to both systems.

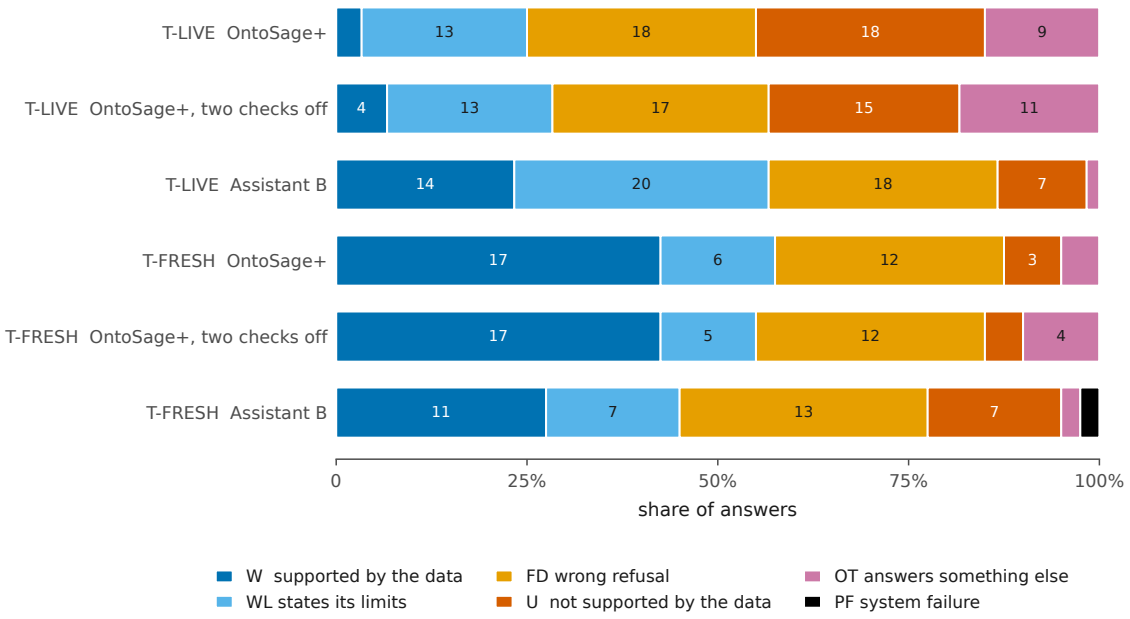


Fig. 30. Labels of the offline answers per system and question set (T-LIVE: the 60 task questions; T-FRESH: 40 own questions, new to both systems). Each bar shows one system's answers to one set as shares; numbers are answer counts.

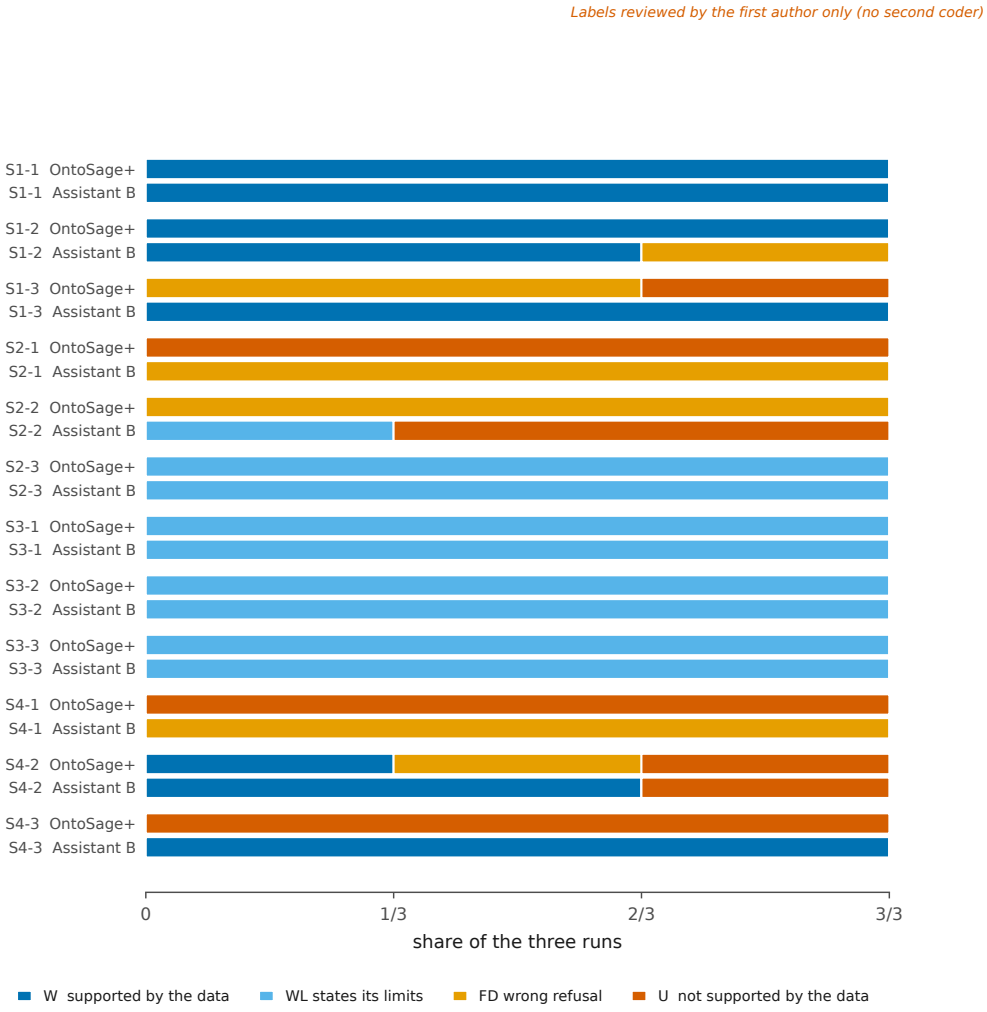


Fig. 31. The twelve canonical task questions, each asked three times of each system: the share of the three runs given each label. The first author reviewed these labels alone, with no second coder.

G What Each Task's Key Rests On

Table 38 lists the twelve study forms from the task bank frozen on 14 October 2026, with the question each asks, the answer its key fixes and what the key rests on. Of these forms, 4 rest at least in part on real readings and the other 8 on none at all. A key can rest on more than one kind of data, so the counts by kind overlap: 4 forms rest at least in part on generated series, 2 on constructed records and 6 on a deliberate absence. The 3 forms of S3 rest only on the absence of what they ask about.

Table 38. What each task's key rests on.

Form	Question (researchers' wording) and key	What the key rests on	Kind of data
S1-1	Did Room 5.02 stay between 20 and 24 °C from 09:00 to 12:00 on Tuesday 14 January 2025? *Key:* Yes	temperature: measured (real snapshot, wide table)	real readings
S1-2	Which had the lower average CO_2 between 09:00 and 17:00 on Tuesday 21 January 2025: Room 5.04 or Room 5.12? *Key:* Room 5.04	CO_2: measured (real snapshot, wide table)	real readings
S1-3	When was the boiler last serviced, and is its next service due before the end of 2026? *Key:* No	service record: constructed register (Planned Maintenance Schedule, version 2026.9)	constructed record
S2-1	Was CO_2 below 1,000 ppm across the whole building on Wednesday 26 February 2025? *Key:* Only Floor 5 measured; whole building not established	CO_2 Floor 5: measured (real snapshot); Floors 0–4: no CO_2 rows on that date (series begin July 2026)	real readings, deliberate absence
S2-2	Could I have used meeting room 2.14 from 08:00 to 09:00 on Monday 20 July 2026? *Key:* Unoccupied (count 0 throughout); booking status not held for that date	occupancy count: generated; bookings: none held for that date	generated series, deliberate absence
S2-3	Can Room 3.68 hold 12 people? *Key:* Capacity not recorded	capacity: none recorded; occupancy count: generated	generated series, deliberate absence
S3-1	What was the air pressure in Room 3.52 on Thursday 3 September 2026? *Key:* Not measured in that room	absent sensor type in the room	deliberate absence
S3-2	When were the walls in Room 0.01 last repainted? *Key:* No redecoration record held	absent record type	deliberate absence
S3-3	What was the temperature in Room 2.40 at noon on Monday 10 August 2026? *Key:* No Room 2.40 in the building	absent referent	deliberate absence
S4-1	Which of Rooms 5.07, 5.14, 5.18 and 5.21 holds at least 8 people and stayed below 25 °C from 14:00 to 15:00 on Friday 21 February 2025? *Key:* Room 5.07	temperature: measured (real snapshot); capacity: building's own estimate, 'not certified'	real readings, constructed record
S4-2	Which of Floors 1 and 3 used more electricity from Monday 14 September 2026 to Sunday 20 September 2026, and do the meter readings cover that whole week? *Key:* Floor 3 recorded more; readings missing ≈21.1 h on both floors	electricity: generated (floor meters)	generated series
S4-3	Did CO_2 in Room 5.58 rise between 09:00 and 13:00 on Wednesday 29 July 2026 while the number of people stayed about the same? *Key:* Yes (descriptive; no cause established)	CO_2: generated; occupancy count: generated	generated series

From the task bank frozen on 14 October 2026: each form's question in the researchers' wording, the answer its key fixes, and its provenance as the bank records it. Kind of data, assigned from that text by rule: a clause naming the real snapshot is real readings; one naming generated data, generated series; one naming a constructed register or the building's own estimate, a constructed record; one stating that nothing is held (none, absent, no), a deliberate absence.

H Sample Questions from the Corpus

Table 39 lists 24 questions we chose to span eight domains, the four question-writing stages and the three complexity labels. Each appears exactly as in the corpus, with the contributor's own spelling, spacing and capitalisation. The whole anonymised corpus is released with the frozen build (Data and Materials). The domain and complexity columns give the automated classifier's labels unchanged, so some differ from what a reader would choose (§3.5).

Table 39. Sample questions from the corpus, printed exactly as contributors typed them. Domain and complexity are the automated classifier's labels (agreement with a second automated classifier in Table 4). Stage: S1 Open Minded, S2 Sensing & Semantics, S3 Scenario-Based, S4 Goal-Oriented. Lkp, Agg, Mlt: LOOKUP, AGGREGATION, MULTI_STEP.

Domain	Stage	Cpx.	Question
Energy	S1	Lkp	are the lights using too much electricity?
	S3	Agg	How much Standby Load do the vending machines draw when the building is closed on weekends?
	S4	Mlt	Could you provide a priority alert to the facilities team if any standby loads exceed a certain threshold during the night?
Air quality	S1	Lkp	are there any sudden drops in air quality anywhere in the building?
	S3	Agg	How much CO2 is in the meeting room?
	S4	Mlt	Does CO2 spike in meeting rooms at 2:00 PM, and is that why everyone feels tired?
Thermal	S1	Lkp	What is the temperature in this room right now ?
	S3	Agg	Can the temperature in the room drop automatically over time, since the room will heat up over the course of the meeting?
	S4	Lkp	how does the building manage airflow to improve comfort?
Lighting	S1	Lkp	Can you turn off the lights in the conference room?
	S2	Lkp	is artificial lighting being overused despite available daylight?
	S4	Mlt	If the Glare Index is too high, can the building suggest a different desk to the user rather than just closing the blinds?
Safety	S1	Lkp	can you guide to the nearest safe exit
	S2	Lkp	Are PM2.5 levels safe inside the building?
	S4	Mlt	Could the building alert a room if pollution levels have risen too high and ventilate until safe to return?
Security	S1	Lkp	Does this Building have onsite round the clock Security Personnel?
	S2	Lkp	Can the sensors then actually close or open doors/windows?
	S3	Agg	When you're tracking visitor count, can you actually track who is visiting and when, without people manually looking at cameras or something?
Occupancy	S1	Lkp	Can I get notifications if amenities are crowded?
	S4	Lkp	Can people in the building receive real time alerts when a workspace is noiser than usual?
	S4	Mlt	Could smart buildings analyze lift occupancy during peak times and then suggest alternative routes to reduce wait times?
No domain matched	S1	Lkp	What happens if a smart system isn't working properly? What do I do?
	S2	Lkp	are there any unusual patterns in movement throughout the building?
	S4	Lkp	are PMV/PPD scored optimal today